

SYSTEM ADMINISTRATION AND MAINTENANCE

Module 1 Reviewer – What is System Administration

1. What is System Administration?

System Administration is the practice of managing, maintaining, and supporting an organization's IT infrastructure to ensure systems run **securely, efficiently, and reliably**.

Key Idea:

System administrators (sysadmins) work mostly in the **background**, preventing IT disasters and ensuring continuity of services.

Scope of Work Depends On:

- Size of the organization
- Complexity of infrastructure

Related Roles:

- System Administrators
- Network Administrators
- Database Administrators

2. Responsibilities of a System Administrator

A system administrator is responsible for **IT services** such as:

- Email services
- File storage
- Websites
- Authentication services
- Network and server uptime

Core Responsibilities:

- Managing servers and services
- User and hardware provisioning
- Routine system maintenance
- Security updates and patching
- Documentation and policy enforcement
- Troubleshooting and issue management
- Vendor communication
- Risk assessment and disaster recovery

3. Understanding Servers

What is a Server?

A **server** is a computer or software that:

- Provides services to other computers (clients)
- Can serve multiple clients at once
- Can be *any* computer

Server Characteristics:

- Usually runs **24/7**
- Designed for reliability and performance

Types of Servers:

- **Tower servers** – stand-alone, upright
- **Rack servers** – flat, mounted in racks
- **Blade servers** – slim, high-density servers

4. KVM (Keyboard, Video, Mouse)

KVM Switch allows:

- One keyboard, mouse, and monitor
- To control multiple computers/servers

Useful for managing many servers efficiently

5. Cloud Computing

Definition:

Cloud computing allows access to:

- Applications
- Files
- Data
- Services from anywhere in the world via the internet.

Key Factors:

- Convenience
- Accessibility
- Cost considerations
- Dependency on internet/service provider

6. Data Centers

A **data center** is a facility that houses:

- Hundreds or thousands of servers

Examples:

- Large companies (Google, Facebook): own data centers
- Smaller companies: **rent space** (colocation / cloud)

7. Organizational Policies

Policies define **acceptable IT usage**, such as:

- Can users install software?
- Password requirements (8+ characters, symbols, numbers)
- Internet access (social media, non-work sites)
- Security settings on company devices

Sysadmins document policies and store them in:

- Internal wikis
- File servers
- Documentation tools

8. IT Infrastructure Services

Sysadmins manage services by:

- Setting them up/Keeping them updated
- Patching security vulnerabilities
- Ensuring hardware/software compatibility

Management is **ongoing**, not one-time setup.

9. User and Hardware Provisioning

User Management:

- Create user accounts
- Assign permissions
- Remove users when they leave

Hardware Management:

- Ensure machines function properly
- Provide necessary software
- Standardize configurations

Hardware Lifecycle:

1. **Procurement** – purchasing hardware
2. **Deployment** – setting up for users
3. **Maintenance** – updates and repairs
4. **Retirement** – secure disposal or reuse

10. Routine Maintenance & Updates

Best Practices:

- Regular security patching
- Batch updates (e.g., monthly updates)
- Keeping systems secure and up to date

Failure to update = high security risk

11. Vendors and Equipment

Sysadmins work with vendors for:

- Printers
- Phones
- Servers
- Conferencing tools

Examples of vendors:

- Dell
- HP
- Apple

Business accounts often provide **discounts**

12. Troubleshooting and Issue Management

Troubleshooting Steps:

1. Ask questions
2. Isolate the problem
3. Follow logs and error messages
4. Identify root cause

Tools:

- Ticketing / bug tracking systems

Soft Skills:

- Customer service

- Empathy
- Proper communication
- Handling difficult users

13. Backup and Disaster Recovery

Key reality:

Something will always break eventually.

Preparedness Includes:

- Regular data backups
- Recovery plans
- Minimizing data loss

14. Applying Changes Safely

Administrator Rights:

- Use admin privileges only when necessary
- Minimize time in admin sessions
- In Linux: use `sudo`

Best Practices:

Respect privacy

- Think before executing commands
- Document all actions

“With great power comes great responsibility.”

15. Rollback and Testing

Rollback:

- Reverting systems to a previous state

Production Environment:

- Live systems used by real users

Never test in production

Test Environment:

- Virtual machine or standby system
- Same configuration as production
- No real users affected

Canary Deployment:

- Deploy changes to a **small group of servers first**
- Observe results before full rollout

16. Risk Assessment

The higher the:

- Number of users affected
- Importance of the service

The more careful and tested changes must be

17. Fixing Issues the Right Way

Proper Fixing Process:

1. Identify steps that led to the issue
2. Define expected vs actual behavior
3. Reproduce the problem in testing
4. Apply fix
5. Re-test
6. Deploy safely (not in production first)

Module 2 Reviewer – Infrastructure Services

1. What are IT Infrastructure Services?

IT Infrastructure Services are the **core services that allow an organization to operate** and deliver IT functionality.

These services enable:

- Internet connectivity
- Internal networking
- Communication between systems
- Hosting applications and data
- Centralized management of users and devices

Without infrastructure services, an organization **cannot function efficiently**.

2. Role of IT Infrastructure Services

The role of infrastructure services is to:

- Support daily business operations
- Ensure availability, reliability, and security of systems
- Allow users to access resources seamlessly

Organizational Size Matters:

- **Small companies:** one admin handles everything
- **Large companies:** teams specialize in specific services

3. Types of Infrastructure Services (Cloud & On-Prem)

3.1 Infrastructure as a Service (IaaS)

- Provides **virtual machines**
- Behaves like a physical server
- You control OS and services

Example: Virtual servers in cloud platforms

3.2 Networking as a Service (NaaS)

- Outsources networking hardware and services
- Reduces cost of expensive networking equipment

Example: Cloud-managed networking

3.3 Software as a Service (SaaS)

- Software accessed via browser
- No local installation required

Examples:

- Microsoft 365
- Google Workspace (G Suite)

3.4 Platform as a Service (PaaS)

- Full platform to:
- Build applications
- Store data
- Run services
- Abstracts infrastructure management

Developers focus on coding, not hardware

3.5 Directory as a Service (DaaS)

- Centralized management of:
 - Users
 - Computers

Enables adding, updating, and removing users easily

Example: Active Directory–style services

4. Server Operating Systems

Server OS are optimized to handle server workloads.

Examples:

- **Windows Server**
- **Linux Server** (Ubuntu Server)
- **macOS Server**

Characteristics:

- More secure than desktop OS
- Built-in server services
- Designed for reliability and uptime

5. Dedicated vs Virtualized Services

Dedicated Hardware:

- One service per physical server
- Better performance
- Higher cost

Virtualized Services:

- Multiple virtual machines on one server
- Lower cost
- Easier maintenance and migration

Key Comparison Factors:

- **Performance** – Dedicated is faster
- **Cost** – Virtualization is cheaper
- **Maintenance** – Virtualized services are easier to manage
- **Failure** – Redundancy and backups are critical

6. Network Services

6.1 File Transfer Services

FTP (File Transfer Protocol)

- Legacy protocol
- No encryption
- Used mainly for web content

Not secure

SFTP (Secure File Transfer Protocol)

- Secure version of FTP
- Uses SSH
- Encrypted data transfer

Preferred over FTP

TFTP (Trivial File Transfer Protocol)

- No authentication
- Used for:

- ☐ OS installation files
- ☐ Network booting

Used with PXE (**Preboot Execution Environment**)

7. Network Time Protocol (NTP)

NTP synchronizes clocks across networked machines.

Why important:

- Log accuracy
- Security auditing
- Consistent system behavior

8. Intranet and Proxy Services

Intranet:

- Private internal company network
- Used for:

- ☐ Documentation
- ☐ Announcements
- ☐ Collaboration

Proxy Server:

- Acts as an intermediary between:
 - ☐ Internal network
 - ☐ Internet

Benefits:

- Privacy
- Security
- Traffic control

9. DNS (Domain Name System)

What DNS Does:

- Translates **domain names** → **IP addresses**

Example:

google.com → 142.250.xxx.xxx

Why Organizations Run Their Own DNS:

- Host websites
- Control internal name resolution
- Enable remote access
- Manage internal services

Essential for web servers and internal networks

10. DHCP (Dynamic Host Configuration Protocol)

DHCP automatically assigns:

- IP address
- Subnet mask
- Default gateway
- DNS server

Eliminates manual IP configuration

Reduces errors and administrative work

QUICK COMPARISON TABLE (MEMORIZE)

Service	Purpose
FTP	File transfer (not secure)
SFTP	Secure file transfer
TFTP	Network boot / installation
NTP	Time synchronization
DNS	Name to IP mapping
DHCP	Automatic IP assignment